

CHE 205 HW 1
Due: 11.59 pm on 01/26/2020

1. (20 points)

For the following functions, graph the function and then use Goal Seek to find the roots(s).

a. $f(x) = x - x^{1/3} - 2$

b. $f(x) = x \tan x - 1$

c. $f(x) = x^4 - e^x + 1$

d. $f(x) = x^2 e^x - 1$

2. (20 points)

Find the minimum and maximum of this function in the range of x is equal to or bigger than -10 and equal to or smaller than 10. Also find the minimum and maximum values when x is positive number.

$$f(x) = e^{-0.3x} \cdot \sin(x+1)$$

3. (30 points)

You are working as a design engineer for a brewery which produces two different types of beer (CHE Yellow and CHE Orange). Each Barrel of CHE Yellow requires 1 oz of hops, 25 pounds of malt. CHE Orange needs 25 pounds of corn, 2 ozs. of hops and 8 pounds of malt. The profit for CHE Yellow is \$30 per barrel and CHE Orange is \$20 per barrel. You need decide how many barrels of each type, brewery need to produce to maximize the profit. The brewery has 500 pounds of corn, 60ozs. of hops and 900 pounds of malt on site.

4. (30 points)

You are working as a process engineer for a gas processing company. Company receives fixed amount of raw gas every week and converts it to two grades of heating gas: regular and premium quality. Regular gas yields in \$125 per metric ton and premium yields \$150 per metric ton. Because of safety concerns, plant can only store 9 metric tons of regular gas and 7 metric tons of premium gas. Also, based on material constraints, summation of 7 times regular gas and 11 times premium can not exceed 77 metric tons. Workers schedule allow working 10 times regular plus 8 times premium needs to be less than or equal to 80. With these limitations, you need to design plant output while maximizing the profit.